

Global Tracking Compatibility Review — GCN Cargo Container Token

TESTNET / DEVNET ONLY · PRE-AUDIT · PROOF-OF-CONCEPT · NOT DEPLOYED FOR LIVE VALUE

Scope	Can the GCN container token integrate with global container tracking systems and the largest ocean carriers?
Verdict	Now yes at the data layer. No at the operational layer — the formats are built and tested; no live feed is connected.
Status	Testnet / devnet only · pre-audit · proof-of-concept · not deployed for live value
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1. Headline findings

Three things were missing that would have blocked carrier integration outright. All three are now built and tested.

- No carrier-standard event format.** The ledger spoke only its own schema. The world's largest carriers publish **DCSA Track & Trace**; terminals send **UN/EDIFACT CODECO** and carriers send **IFTSTA**. Without these, integration meant a bespoke adapter per partner.
- No cross-industry visibility format.** Shippers, retailers and customs systems consume **GS1 EPCIS 2.0**, not shipping-specific formats. Walmart and others mandate EPCIS 2.0 from suppliers.
- Unvalidated industry identifiers.** IMO numbers were checked for length only (they have a check digit), MMSI was not validated at all, and the size/type field held yard shorthand that carrier systems reject.

One finding changes a stated plan:

FleetMon's API is retired. FleetMon and MarineTraffic were both acquired by Kpler in February 2023, and **FleetMon was phased out from January 2024** as the two merged. The developer portal and `apiv2.fleetmon.com` are gone. Building the AIS integration on FleetMon means building on something that is no longer there.

The adapter was therefore written **provider-agnostic**: a normalised position shape with per-provider mappings. MarineTraffic/Kpler is the live path, **generic** takes a raw AIS feed or any other vendor, and the FleetMon profile is retained only for archived exports and legacy contracts. Switching provider is one argument, not a rewrite. The AIS vendor decision is now reversible, which given the consolidation in this market is worth more than any single integration.

2. What was built in response

Component	What it does	Tests
<code>shared/identifiers.js</code>	IMO check digit (ISO 8713), MMSI + MID with station-type detection, ISO 6346 size/type codes, SCAC, SMDG/BIC facility codes, DCSA document references	4
<code>standards/dcsa.js</code>	DCSA Track & Trace 2.1 events ↔ GCN waypoints, both directions	9
<code>standards/edifact.js</code>	CODECO and IFTSTA parser → GCN waypoints, with UNA service-character support	6
<code>standards/epcis.js</code>	GCN journey → GS1 EPCIS 2.0 JSON-LD with CBV business steps and GIAI identifiers	4
<code>standards/ais.js</code>	Provider-agnostic AIS adapter (MarineTraffic / generic / FleetMon-legacy) with voyage-window enforcement and track thinning	9

32 interoperability checks passing, on top of the 19 EVM, 14 geodesy and 15 agent tests already in the module. **80 automated checks in total, all passing.**

3. Three refusals built into the adapters

These are the design decisions that matter most, and each one is enforced in code and covered by a test. They all cut the same way: the ledger would look better without them, and be worth less.

3.1 Forecasts never become attested positions

Most of the event volume on a carrier feed is **EST** (estimated) or **PLN** (planned), not **ACT** (actual). Both the DCSA adapter and the EDIFACT parser accept only actuals and return the rest as visible rejections.

An ETA written into an append-only ledger becomes indistinguishable from an observation the moment it is stored. Carriers revise ETAs constantly; the ledger would fill with positions the container never occupied, and nothing downstream could tell which was which.

3.2 A ship's position is not a container's position

The AIS adapter derives **voyage windows** from the container's own `LOADED_VESSEL` → `DISCHARGED` events, and emits a container waypoint only for AIS positions that fall inside one, on the matching IMO. Everything else is rejected with: *"This is the ship's position, not the container's."*

This is the single most common way container tracking misleads people. A vessel's transponder keeps reporting after the box is on a truck, and a naive integration keeps attributing that track to the container. With no `LOADED_VESSEL` event at all, the adapter refuses every position for that container.

3.3 A UN/LOCODE is a port, not a fix

DCSA and EDIFACT events carry a location code, not coordinates. The resolved position is a **port centroid, accurate to kilometres**. Every waypoint from those adapters carries `positionBasis` saying so, so it is never presented as a GPS fix.

4. Compatibility matrix

Standard / system	Status	Note
ISO 6346 container identity	✔ Built & tested	Check digit validated; one token per id
ISO 6346 size/type codes	✔ Built & tested	Now enforced on-chain ; yard shorthand rejected
IMO ship number (ISO 8713)	✔ Built & tested	Check digit validated
MMSI (ITU-R M.1371)	✔ Built & tested	MID range + non-ship-station detection
DCSA Track & Trace 2.1	✔ Built & tested	Bidirectional; ACT-only
UN/EDIFACT CODECO	▲ Common subset	Per-partner MIG validation required
UN/EDIFACT IFTSTA	▲ Common subset	Same caveat
GS1 EPCIS 2.0	✔ Built & tested	JSON-LD, CBV steps, GIAL, sensor elements
UN/LOCODE	▲ Demo table	~22 hand-typed centroids; needs the official UNECE list
SMDG facility codes	▲ Shape only	Codes are partner-assigned
SCAC	▲ Convenience table	14 major carriers; verify against NMFTA
AIS providers	✔ Built & tested	MarineTraffic/Kpler, generic, FleetMon-legacy
AIS — live feed	✖ Not connected	No credentials, never run against a live API
Container IoT telemetry	✖ Not built	Traxens, Nexxiot, ORBCOMM, Globe Tracker — the only TIER1 source
UN/EDIFACT BAPLIE	✖ Not built	Stowage plan; needed only for on-vessel position
WCO Data Model / customs	✖ Not built	Needed for customs filing, not for tracking
ISO 18185 e-seals	✖ Not built	Seal numbers parsed from CODECO; no e-seal reader

Legend: ✔ built and tested offline · ▲ built with a stated limitation · ✖ not built

5. What "compatible with the largest shippers" honestly means right now

The carriers behind DCSA — Maersk, MSC, CMA CGM, Hapag-Lloyd, ONE, Evergreen, HMM, Yang Ming, ZIM — publish Track & Trace APIs against the standard this module now reads and writes. So:

What is true: the data formats are built, bidirectional, and tested against realistic event payloads. A DCSA event from any of those carriers maps into the ledger, and a GCN waypoint maps back out into a DCSA event their systems can consume.

What is not yet true: nothing is connected. No carrier API credential exists, no live feed has been consumed, and each carrier extends and constrains DCSA in its own published API. Integration is a per-carrier commercial and contractual exercise — an API agreement, sandbox credentials, then field-by-field validation against *their* documentation. The code is ready for that conversation; it has not had it.

The honest claim to make externally is: **"DCSA Track & Trace compatible, pending carrier API agreement"** — not "integrated with major carriers."

6. Still missing, in priority order

- Container IoT telemetry (TIER1).** Still the biggest gap, and now the *only* remaining one that changes what the product fundamentally is. Everything built so far improves TIER2 and TIER3 — paperwork and ships. **Nothing in this module yet observes a container.** Traxens, Nexxiot, ORBCOMM and Globe Tracker are the vendors; one instrumented lane demonstrates more than a hundred AIS-derived ones.
- One live carrier feed.** Pick a single carrier, get sandbox credentials, validate the DCSA mapping field-by-field against their published API. Until this happens, "compatible" remains a statement about code, not about a working integration.
- An AIS provider decision.** FleetMon is gone. MarineTraffic/Kpler is the direct successor; Spire and ORBCOMM are the main alternatives. The adapter makes this reversible, but the commercial decision still has to be made.
- The official UN/LOCODE list.** ~22 hand-typed port centroids will not survive contact with a real route. Load the UNECE code list and record the edition.
- Per-partner EDIFACT profiles.** The parser handles the common subset. Each terminal's MIG differs; profiles should be configuration, not code changes.
- BAPLIE** if on-vessel stowage position is ever needed (bay/row/tier).
- Audit** before any mainnet consideration. Unchanged and non-negotiable.

7. Changes made to existing code

- `ContainerToken.sol` now rejects yard shorthand in the size/type field. ISO 6346 requires the 4th character to be a digit (`45G1`); shorthand like `40HC` is also four characters but ends in a letter, so the digit check is what separates them. This was a live interoperability defect: the sample data carried `40HC`, which no carrier system accepts.

- **Sample data and all tests** updated from `40HC` to `45G1`.
 - Contract recompiled, **19 EVM tests passing**.
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8. Findings from verification worth recording

Two defects were caught by the tests themselves rather than by review:

- The EDIFACT parser rejected a container id in my own test fixture. `MSCU1234565` has a bad check digit — it should be `MSCU1234566`. The validator was right and the fixture was wrong, which is the correct direction for that disagreement.
 - The on-chain size/type check initially tested length only. `40HC` is four characters, so it passed. The test failed, and the rule was corrected to the actual ISO 6346 structure.
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Sources

- DCSA Interface Standard for Track and Trace 2.1 — <https://dcsa.org>
- DCSA event code reference — <https://docs.vizionapi.com/docs/dcsa-event-code-types>
- UN/EDIFACT CODECO D95B — <https://www.stylusstudio.com/edifact/D95B/CODECO.htm>
- GS1 EPCIS 2.0 — <https://www.gs1.org/epcis>
- FleetMon / MarineTraffic merge and phase-out — <https://support.marinetraffic.com/en/articles/9552991-fleetmon-and-marinetraffic-merge-process-for-former-fleetmon-ais-partners-from-january-2024>
- FleetMon API retirement and historical surface — <https://github.com/api-evangelist/fleetmon>
- Pole Star Maritime Transparency Index — <https://www.polestarglobal.com/maritime-transparency-index/>